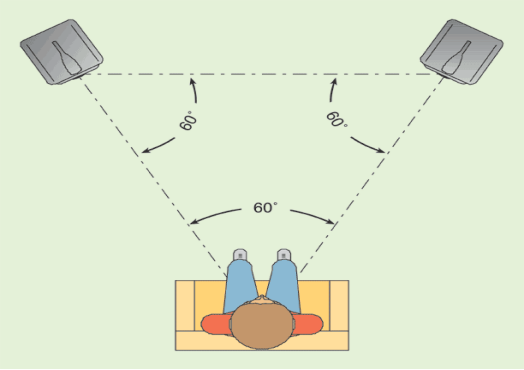
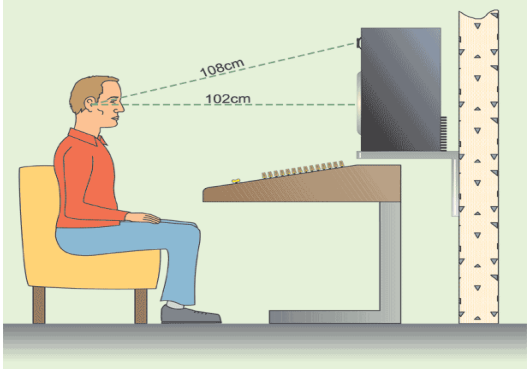
The level-meter is going to be needed since we have a room to monitor our sounds. The purpose of a level-meter is to alert you to excessive volume levels and for frequency response at your ‘sweet spot’.

The fifth subject is placing your speakers in the corners of your room. Why? Some problems with boosting the bass, resulting in inaccuracies. The ideal speaker position will place the listener and speakers at points on an equilateral triangle, as shown in the diagram below.





Courtesy: Sound on Sound Magazine.

Placing the speaker symmetrical is very important because walls, floors, and ceiling interact with speakers.

THE NEAR-FIELD MONITOR

We do not have to confuse ourselves with the monitor of a computer screen since they share the same name.

When you visit a professional studio, you will notice that large monitors have been mounted at considerable distances, around six to ten feet away from the mixing engineer. On the contrary in home or budget studios, near-field monitors have become the norm to monitor. With this technique, small but accurate and efficient speakers are placed from three to six feet from the mixer’s ears, with the head and speakers forming a triangle (equilateral). The speakers should point toward the ears and be at ear level.

Benefits of using near-field monitors:

- They minimize the impact of room acoustics on the overall sound, as the speakers' direct sound is far greater than the reflections coming off the room surfaces.
- They do not need to produce a lot of sound (volume) because of proximity (closeness) to your ears.
- Closer also means relaxation of the requirements of the amplifier.

PASSIVE OR ACTIVE MONITORS

There are two main types of monitors: the active monitor and the passive monitor.

- **Active monitors** incorporate the amplification needed to drive the speakers from a line-level signal.
- **A passive monitor** consists of only the speaker and crossover, so they require an outboard amplifier.

HOW TO CHOOSE A MONITOR

It is recommended to go for active near-field monitors due to their reliability and durability. Here are some tips for choosing a monitor.

1. Go for the monitor that best suits your listening space and imperfect hearing response. That is, choose the part that colors your sound the way you prefer.
2. Take along a couple of your cds and try several monitors to capture the preferable listening experience.
3. Go for magnetically shielded speakers since most of our computer screens would be placed between them. They will prevent any monitor distortions of your computer screen.

Reflections

One of the realities in audio production is that what we hear when listening to our song through a pair of monitor loudspeakers is **a combination of the direct sounds (on-axis), after it has reflected from the walls, corners, and other objects within our rooms**. These two combinations unfortunately create illusions of low frequencies (bass) and every sound we hear is inaccurate. But there is good news for budget studio owners. Below are some practical solutions.

Solution 1

Evidently, small rooms cannot generate very deep bass so the best option is to acquire monitors with a flat response down to around 60 – 70 Hz (this information can be located at the back of the monitors) and rolls off gradually towards the low-end of the spectrum. At this point, the room will not affect the bass-end quite as much.

Solution 2

The positioning of the monitors is crucial to eliminating inaccuracies caused by the on-axis and off-axis effect. The distance between the speakers and the nearer surfaces should ideally be as random as possible, to stop all of the reflections combining at the same frequency.

Rule 1

Keep the speaker at least a foot from the wall in front of you, and at least 18 inches from the sidewalls.

Rule 2

In rectangular rooms, it is wise to keep the speakers far away from the corners and sidewalls as possible. This reduces the extent of reflections between the listener and the walls.

Rule 3

The speakers should be erected on stands a little way behind the desk.

Rule 4

The tweeters must be pointed towards the head of the listener.

Rule 5

Ensure that the speakers do not rattle or vibrate on the stands.

MONITORING AND ROOM ACOUSTICS

One will be puzzled to know that the perfect recording environment is the great outdoors. Since that is not possible, the only suitable thing to do is to acoustically treat our home studios. However, we must understand the concept of sound waves and room acoustics.

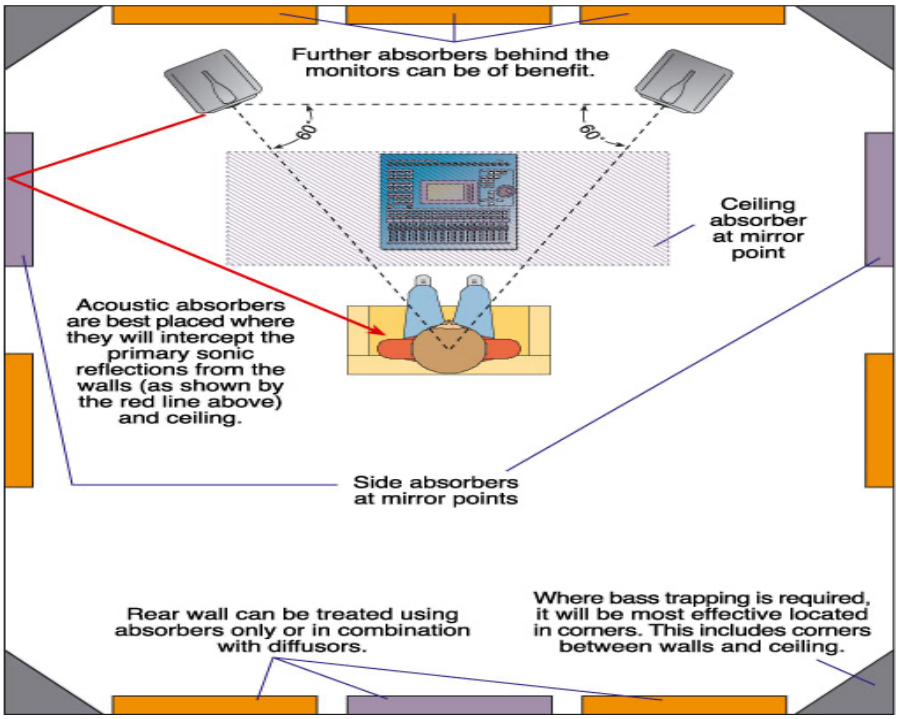


Image Source: Sound on Sound Magazine, All Rights Reserved

The diagram above explains how to apply basic acoustic treatment to a typical home studio room. The absorber panels shown in purple are the most important but adding in the orange absorbers would improve the situation further. Acoustic foam is a common choice of absorber in this application. If bass trapping is required, then it is usually most effectively applied in the room corners (including those corners between any of the walls and the ceiling). Note also the angles and positions of the monitors in comparison to the listening position — arranging the three points in an equilateral triangle will help give a natural stereo image.

Sound waves produced from your monitors radiate out to the borders of your studio, are then reflected, and interact with each other. This can be like the ripple effect. The desired result is not pleasant, so to speak.

Three types of sound reflection theories exist. They are known as **axial, tangential, and oblique modes.**

AXIAL REFLECTIONS

This mode means sound is being reflected from wall to opposing wall or floor to ceiling, or two parallel surfaces.

TANGENTIAL REFLECTIONS

Corners trap bass, thus creating an illusion that the bass is adequate.

OBLIQUE REFLECTIONS

This type of reflection relates to which direction in a room sound is being reflected from one hard surface to another.

Acoustic treatment has become a multi-million-dollar business enterprise. But as far as home or budget studio owners are concerned, the minimum can still make wonders.

A solution exists to achieve the minimum acceptable acoustic levels, by **absorption and diffusion**.

ABSORPTION

Sound played in an acoustically dead environment will bounce off the wall-to-wall or from ceiling to floor or vice versa. Absorption materials such as acoustic foams absorb sound so that the listener only listens to what he/she is playing, instead of fluttering and echoes and so on.

The two latest materials for acoustic treatments are high-quality acoustic foams (studio foams) and specialized acoustic fiberglass.

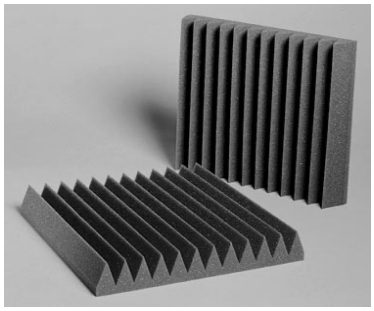


Image Source: Auralex Acoustics. All Rights Reserved.

Some benefits of acoustic foams include the following:

- Ready to work with
- Simple to trim to size, and

- Cost-effective for any type of budget studio
- Improves the sound.
- Produces accurate monitoring.
- Recordings sound better.
- What you hear is what you get.
- Creates a productive and comfortable environment.
- Space is enjoyed.
- Improved concentration and hearing acuity.

NOTE. BASED ON PHYSICS, THE THICKER THE FOAM AND THE GREATER THE AMOUNT OF OVERALL ABSORPTION

DIFFUSION

Acoustic sound diffusion prevents sound waves from grouping, thus eliminating hot spots or nulls in a room. Further, diffusion widens the 'sweet spot' and produces a song, 3-dimensional sense of openness in a room, rendering it easier to hear into a mix.

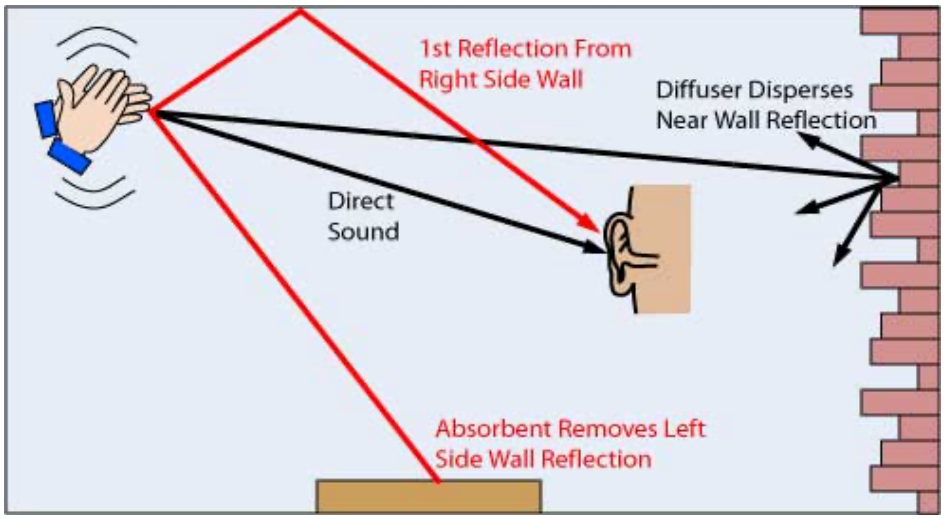


Image Source: Auralex Acoustics

In addition, diffusion obliterates standing waves and flutter echoes without just eliminating acoustic energy from space.

Acoustic sound diffusion also makes a room, smaller in size, seem large and a larger space even larger.

HOW DIFFUSION WORKS AND PERFORMS

Basically, the irregular surface contours and varying angles of the diffuser each reflect sound waves in specific different directions. Sound is simply and accurately reflected at different times.

Thus, a well-designed diffuser spreads sound symmetrically over the sonic landscape in front of it. Sound is re-distributed in the mid and high frequency evenly throughout space, and not to absorb it.

Chords supply harmony and this is how musicians have composed, arranged, and improvised their songs.

In this instance, we can delve deeper into chords. [18]

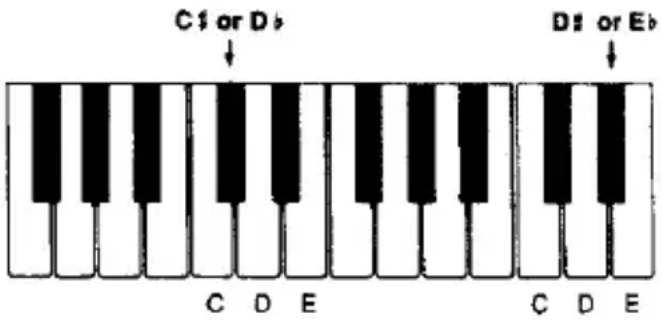


Image Source: How to play popular piano in 10 easy lessons.

Notice that the black keys come in sets of two's and three's.

CONSTRUCTION OF CHORDS

Chords make up the foundation of present-day music. So, what is a chord?

- **A chord is when you play three or more different musical notes at the same time.**
- A chord, to be simpler, a major, or a minor, contains three notes and is called **a triad**, TRI from Latin. [19]
- Moreover, chord symbols assist a keyboard or piano player or any musician to decipher what note to play without forcing them to read sheet music.
- Chord symbols have always been shown in the notation above the treble clef. They may look like a capital letter, sometimes displayed with sharps # or flats *b*